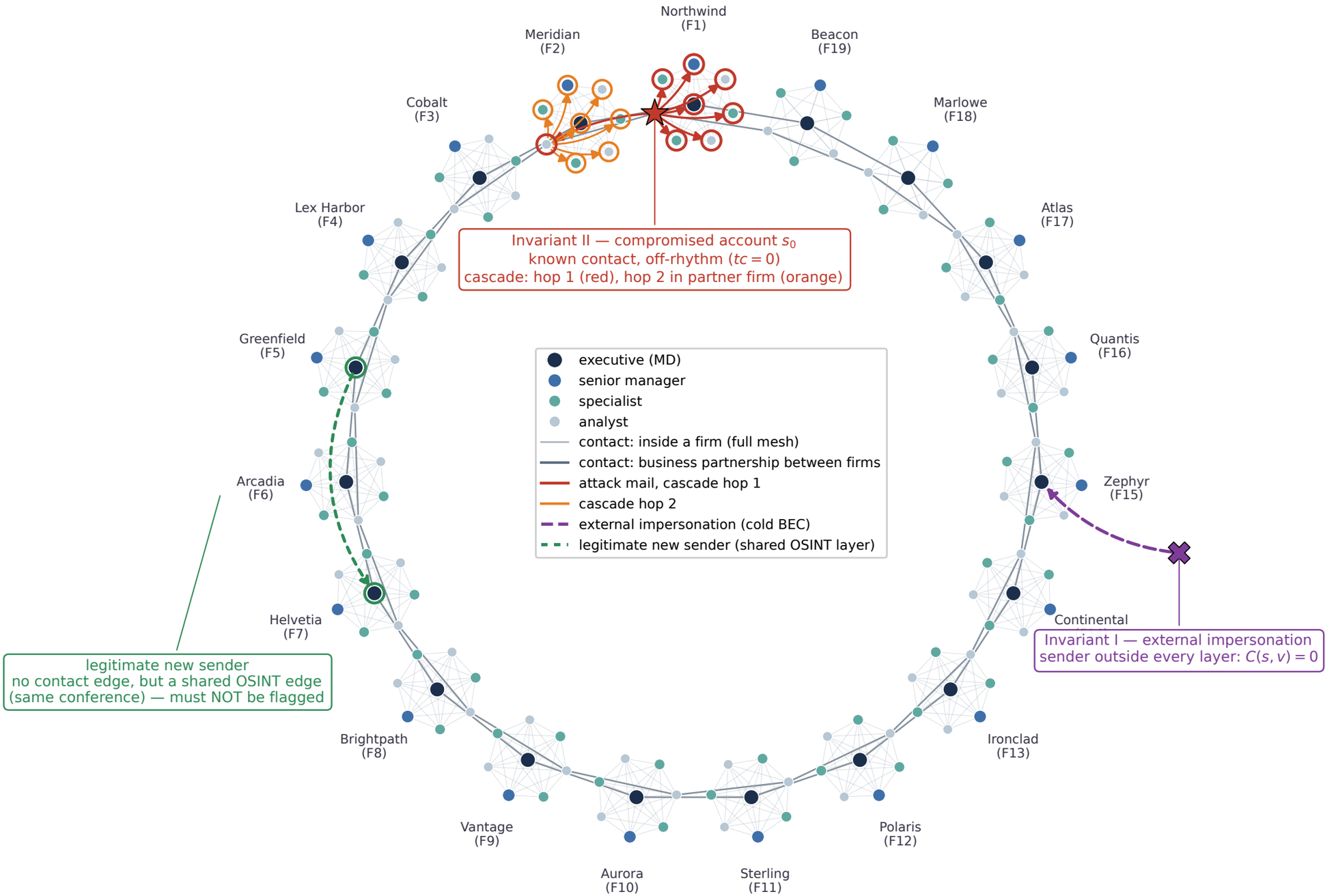
Case-study organization: 19 firms, 150 accounts, 556 contact edges (518 inside firms, 38 between partner firms)
Northwind (F1)
Meridian (F2)
Beacon (F19)
Cobalt (F3)
Marlowe (F18)
Lex Harbor (F4)
Atlas (F17)
Greenfield (F5)
Quantis (F16)
Arcadia (F6)
Zephyr (F15)
Helvetia (F7)
Continental
Brightpath (F8)
Ironclad (F13)
Vantage (F9)
Polaris (F12)
Aurora (F10)
Sterling (F11)
Invariant II — compromised account s_0
known contact, off-rhythm ($tc = 0$)
cascade: hop 1 (red), hop 2 in partner firm (orange)
executive (MD)
senior manager
specialist
analyst
contact: inside a firm (full mesh)
contact: business partnership between firms
attack mail, cascade hop 1
cascade hop 2
external impersonation (cold BEC)
legitimate new sender (shared OSINT layer)
Invariant I — external impersonation
sender outside every layer: $C(s, v) = 0$
legitimate new sender
no contact edge, but a shared OSINT edge
(same conference) — must NOT be flagged